PROJECT REPORT

PROJECT DOCUMENTATION

PROJECT TITLE	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer (ServiceNow Incident Management Project)
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DEPARTMENT	B.Tech Information Technology
COLLEGE	Sri Ramanujar Engineering College
ACADEMIC YEAR	2026 - 2027
GUIDE	

CERTIFICATE

This is to certify that the project report entitled **"Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer"** is a bonafide record of the work carried out by **Madesh M (Team Leader)** and team members of the Department of Information Technology, Sri Ramanujar Engineering College, under the guidance and supervision of the faculty guide during the academic year 2026-2027.

Project Guide

Signature:

Date:

Head of the Department

Signature:

Date:

DECLARATION

We hereby declare that the project work presented in this report is our original work carried out as part of the academic requirements. The work has been completed with proper acknowledgement of the sources, tools, datasets, libraries and references used.

Place:

Date:

Team Members' Signatures

1. Madesh M (Team Leader)

ABSTRACT

In many organisations, requests for standard laptops are still handled through emails, spreadsheets and manual follow-ups. This causes delays, missing information, repeated approvals for routine items and poor visibility of request status for employees and procurement staff.

This project automates the ordering of standard laptops on the ServiceNow platform using **Flow Designer**. A catalog item, *Standard Laptop Request*, collects the required details from the employee. A flow is triggered when the request is submitted. Requests for approved standard models are validated and approved automatically, while non-standard requests are routed to the line manager. The flow then creates a fulfilment task for the IT Procurement group, sends email notifications at key stages and closes the request when the task is completed.

The solution was built on a ServiceNow Personal Developer Instance with low-code configuration and tested using multiple scenarios. It reduces manual effort and turnaround time, enforces the standard-model policy consistently and provides complete tracking of every laptop order.

Keywords: ServiceNow, Flow Designer, Service Catalog, IT Procurement, Workflow Automation, Low-Code, Laptop Request

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1. INTRODUCTION

ServiceNow is a cloud platform widely used for IT Service Management (ITSM). Its Service Catalog lets employees request standard goods and services through a single portal, and Flow Designer is its low-code tool for automating the steps that follow a request. Together they replace email chains and manual hand-offs with a consistent, trackable process.

IT procurement is a good candidate for automation. Laptop requests are frequent, follow a predictable pattern and usually involve a small list of approved standard models. Handling each one manually wastes time for requesters, approvers and the procurement team.

1.1 Project Overview

This project, **"Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer"**, creates an automated ordering process for standard laptops in ServiceNow. An employee submits the *Standard Laptop Request* catalog item. A Flow Designer flow then validates the request, approves it automatically when the chosen model is on the standard list (or sends it to the manager otherwise), creates a task for the IT Procurement team, notifies the requester and closes the request on completion.

The system is intended for employees who need a laptop, line managers who approve exceptions, and the IT Procurement team that orders and delivers devices. The expected outcome is faster, error-free and fully tracked laptop orders with minimal manual effort.

1.2 Purpose

The purpose of the project is to simplify and speed up routine laptop procurement. The main objectives are:

- To provide a single catalog item for requesting standard laptops.
- To automate approval, task creation, notification and closure using Flow Designer.
- To enforce the standard-model policy and send exceptions for manager approval.
- To give requesters and procurement staff clear visibility of request status.

Compared with email- and spreadsheet-based ordering, the proposed system reduces turnaround time, avoids missing information and removes repetitive manual work.

2. IDEATION PHASE

The team identified the problem by examining how laptop requests are handled in a typical organisation and generated solutions using a problem statement, an empathy map and brainstorming.

2.1 Problem Statement

Target users: employees requesting laptops, line managers and the IT Procurement team.
Current difficulty: Laptop requests arrive by email or phone, details are missing, and every request waits for manual approval and manual task creation.
Cause: There is no structured request form and no automated workflow, even for standard, pre-approved models.
Impact: Long delays, repeated follow-ups, errors in orders and no reliable tracking of request status.

How might we let employees order standard laptops quickly while automatically handling approval, fulfilment and communication?

2.2 Empathy Map Canvas

The empathy map represents the primary user, the employee who needs a new laptop, together with the procurement coordinator who processes the request.

SAYS	THINKS
"I only need a standard laptop, why does it take so long?" "Where is my request now?" "I keep getting asked for the same details."	Routine requests should not need manual approval. The process should be simple and visible. Automation can remove repeated work.
DOES	**FEELS**
Sends emails and follows up. Fills forms with missing details. Procurement staff re-enter data into tasks.	Frustrated by delays and follow-ups. Uncertain about request status. Relieved when the process is automatic and clear.
PAINS	**GAINS**
Slow approvals for standard items. Manual task creation and data entry. No tracking or notifications.	Fast, automatic approval for standard models. Instant task creation and email updates. One place to track every order.

Figure 2.1: Empathy map of the employee / procurement coordinator

2.3 Brainstorming

Several approaches were compared on effort, flexibility and maintainability.

Idea	Description	Decision
A. Email-based process	Continue requesting laptops by email.	Rejected - no tracking, error-prone.
B. Catalog item only	Form without automation; staff process manually.	Rejected - still manual and slow.
C. Business rules / scripts	Code-based automation on request tables.	Rejected - harder to maintain and modify.

Idea	Description	Decision
D. Workflow Editor (legacy)	Older graphical workflow engine.	Rejected - Flow Designer is the current standard.
E. Catalog item + Flow Designer	Low-code flow triggered by the catalog item.	**Selected** - fast to build, visual, easy to change.

Idea E was selected because it uses native, low-code features, is easy for non-developers to read and update, and fully automates the process from request to closure.

3. REQUIREMENT ANALYSIS

This phase identifies what the system must do, who will use it and what is needed to build and test it.

3.1 Customer Journey Map

The journey follows an employee who needs a standard laptop and shows how automation improves each stage.

Stage	Need	Request	Approval	Fulfilment	Closure
Actions	Employee needs a new laptop.	Opens portal and fills the laptop catalog item.	Flow auto-approves standard model or asks manager.	Procurement task is created and the laptop ordered.	Task closed; request completed.
Pain points	Unsure how to request.	Too many fields or missing information.	Waiting for approval.	No updates on progress.	Unclear when it is done.
Emotions	Neutral	Hopeful	Impatient (manual) / Satisfied (auto)	Anxious	Happy
Opportunities	Clear catalog entry.	Simple form with defaults.	Auto-approve standard items.	Email updates at each stage.	Closure notification.

Figure 3.1: Customer journey map

3.2 Solution Requirement

Functional requirements

- FR1: A catalog item shall allow employees to request a laptop by choosing a model and giving a justification.
- FR2: A flow shall trigger automatically on submission of the catalog item.
- FR3: Requests for standard models shall be approved automatically.
- FR4: Non-standard requests shall be sent to the requester's manager for approval.
- FR5: A fulfilment task shall be created and assigned to the IT Procurement group.
- FR6: The requester shall receive email notifications and the request shall close when the task is completed.

Non-functional requirements

- Usability: simple, short form accessible from the portal.
- Performance: flow starts immediately after submission.
- Maintainability: low-code flow that can be edited without scripting.
- Reliability: every execution is logged and can be reviewed.

Hardware / software requirements

- Computer with a modern web browser and internet connection.
- ServiceNow Personal Developer Instance (PDI) with Service Catalog and Flow Designer.
- Administrator access to create catalog items, flows and notifications.

User requirements: test users acting as employee, manager and procurement agent, and a procurement assignment group. **Data requirements:** a list of standard laptop models and sample requests (no external dataset is used).

3.3 Data Flow Diagram

The Level 0 DFD shows the system as one process interacting with the employee, the procurement team and the request records.

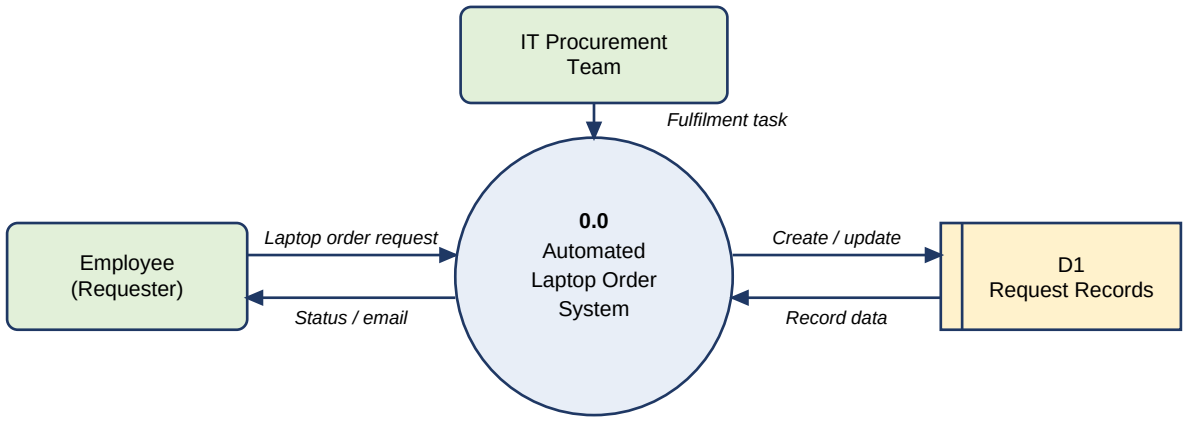

Figure 3.2: Data Flow Diagram - Level 0

The Level 1 DFD has four processes. **1.0 Submit Laptop Request** captures the catalog form. **2.0 Validate & Approve** checks the model, reads user and manager data and approves automatically or requests manager approval. **3.0 Create Fulfilment Task** creates and assigns the catalog task. **4.0 Notify & Close Request** sends emails and closes the request when the task is done.

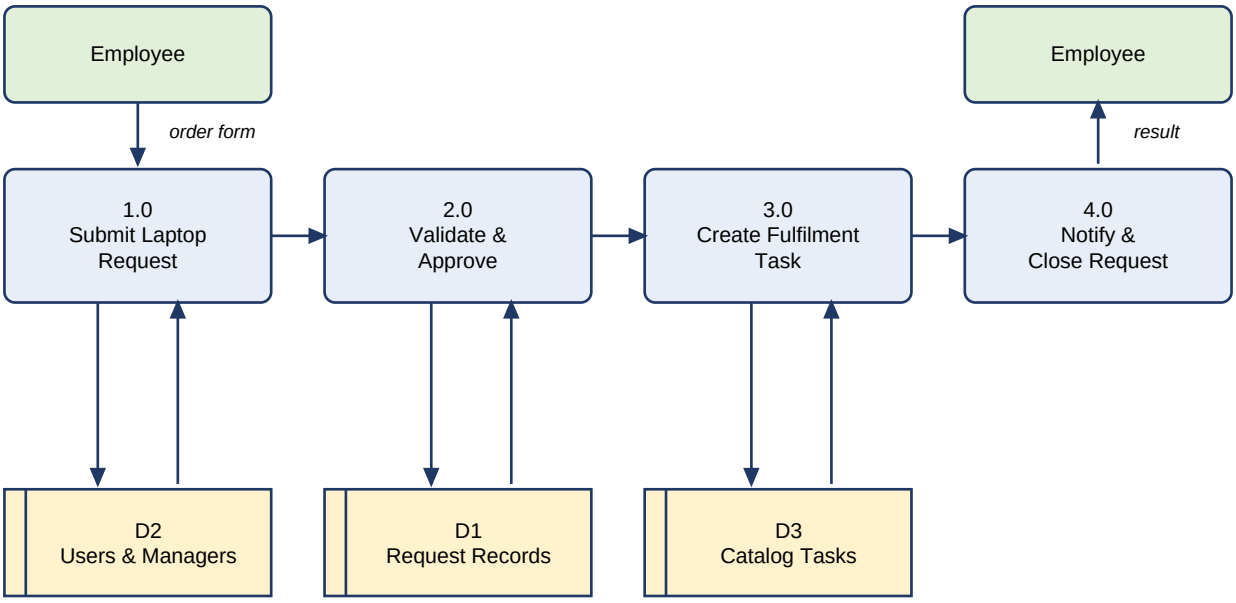

Figure 3.3: Data Flow Diagram - Level 1

3.4 Technology Stack

Layer	Technology	Purpose
Frontend	ServiceNow Service Portal / Employee Center - catalog item form	Employee interface for ordering laptops
Backend	Flow Designer (trigger, actions, flow logic), approvals, notifications	Automation of approval, task creation and closure
Database	ServiceNow tables: sc_request, sc_req_item, sc_task, sys_user, sys_user_group	Data storage
Tools	Personal Developer Instance, Catalog Builder, Flow Designer, Flow Execution view, GitHub	Development, testing and version tracking
Deployment	ServiceNow cloud instance (PDI); transferable using Update Sets	Hosting and migration

4. PROJECT DESIGN

The requirements are converted into a design that uses native ServiceNow features so that the solution is practical and easy to deploy and maintain.

4.1 Problem Solution Fit

User problem	Solution feature	Fit
No structured way to request a laptop	Standard Laptop Request catalog item with variables	Complete and consistent requests
Slow approval for routine models	Flow condition: auto-approve standard models	Instant approval
Exceptions need control	Manager approval for non-standard models	Policy enforced
Manual task creation by procurement	Flow action: create catalog task for IT Procurement	No re-entry of data
No status visibility	Email notifications and request tracking	Transparent progress

4.2 Proposed Solution

The proposed system consists of the following modules and workflow:

- **Catalog module:** create the *Standard Laptop Request* item with variables such as laptop model, requested for and business justification.
- **Flow module:** create a flow in Flow Designer triggered by the Service Catalog.
- **Approval module:** auto-approve standard models, otherwise ask the manager for approval.
- **Fulfilment module:** create a catalog task assigned to the IT Procurement group.
- **Notification module:** email the requester when the order is approved, rejected and completed.

Flow outline:

Step	Flow Designer element	Description
Trigger	Service Catalog	Runs when Standard Laptop Request is submitted.
1	Action: Get Catalog Variables	Reads model, requested for and justification.
2	Flow logic: If	Is the model on the standard list?
3a	Yes: Update Record	Set approval to Approved automatically.
3b	No: Ask for Approval	Manager approves or rejects the request.
4	Action: Create Catalog Task	Assign task to IT Procurement group.
5	Action: Send Email	Notify requester that the order is in progress.
6	Wait for Condition	Wait until the catalog task is closed complete.
7	Update Record + Send Email	Close the request item and notify the requester.

4.3 Solution Architecture

The architecture has three layers. The **presentation layer** is the portal where employees fill the catalog item. The **application layer** contains the Service Catalog application, the Flow Designer automation flow, and the approval and notification engine. The **data layer** stores requests, request items, tasks, users and groups. Flow Designer connects the layers by reading and updating records at each step.

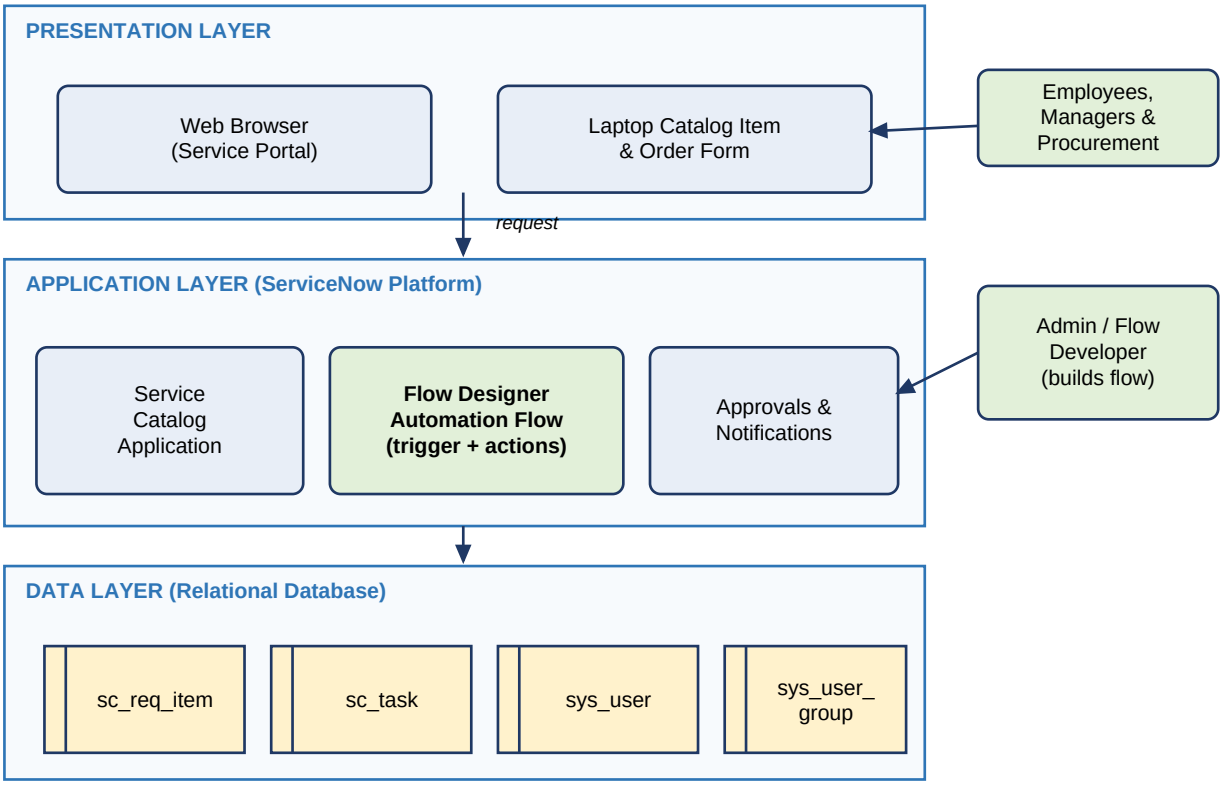

Figure 4.1: Solution architecture

5. PROJECT PLANNING & SCHEDULING

The work was divided into phases with clear responsibilities over an eight-week period.

5.1 Project Planning

Phase	Activities	Weeks	Responsibility
Problem identification	Study manual laptop ordering; problem statement	1	Madesh M
Requirement analysis	Requirements, DFD, standard model list	1-2	Madesh M & team
Design	Catalog item design, flow logic, architecture	2-3	Madesh M
Development	PDI setup, catalog item, flow, notifications	3-5	Team
Testing	Test scenarios, flow execution review, fixes	5-6	Team
Documentation	Report, diagrams, screenshots	6-7	Madesh M
Demonstration	Final demo and submission	8	Team

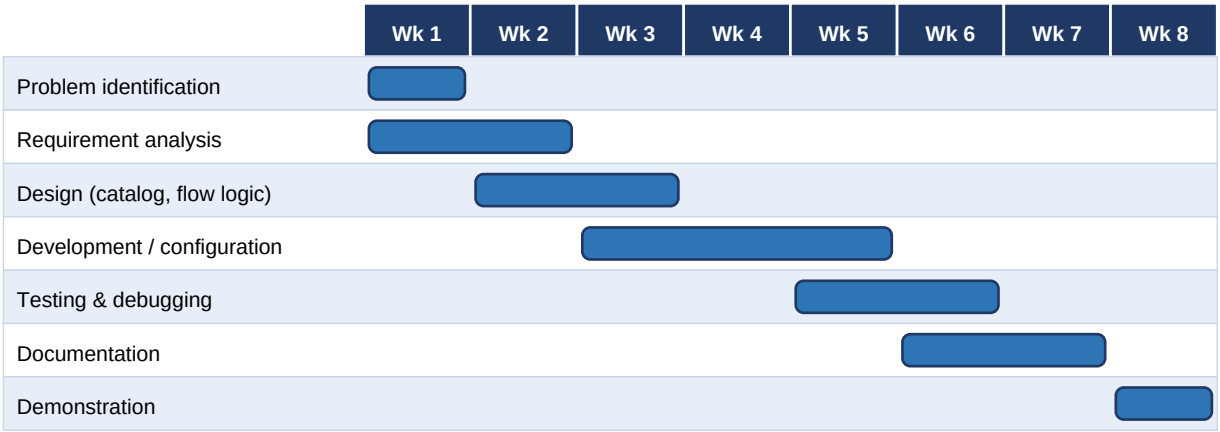

Figure 5.1: Project schedule (Gantt chart)

6. FUNCTIONAL AND PERFORMANCE TESTING

Testing was carried out on the ServiceNow instance by submitting requests as different users and checking the flow execution details.

6.1 Performance Testing

ID	Feature / Scenario	Input	Expected Output	Actual Output	Status
TC-01	Standard model request	Select standard laptop model	Auto-approved	Approved automatically	PASS
TC-02	Fulfilment task creation	Approved request	Task created for IT Procurement	Task created and assigned	PASS
TC-03	Requester notification	Request approved	Email sent to requester	Email received	PASS
TC-04	Non-standard model request	Select non-standard model	Manager approval requested	Approval sent to manager	PASS
TC-05	Manager rejects request	Manager rejects	Request closed; requester notified	Closed and notified	PASS
TC-06	Task completion	Close catalog task complete	Request item closed complete	Request closed	PASS
TC-07	Mandatory variable check	Submit without justification	Form blocks submission	Submission blocked	PASS
TC-08	Flow execution status	Open Flow Execution view	Execution shows Completed	Completed	PASS

Observations: The flow started immediately after each submission and completed without manual intervention for standard models. Notifications were delivered at each stage and every execution was visible in the Flow Execution view, which helps in troubleshooting. The form was short and easy to use. Approval wait time for non-standard requests depends on the manager.

7. RESULTS

The final system allows an employee to order a standard laptop from the portal and have it approved, assigned for fulfilment, communicated and closed automatically. The following screenshots demonstrate the working features.

7.1 Output Screenshots

[Paste screenshot here]

Catalog item form

Figure 7.1: Standard Laptop Request catalog item

[Paste screenshot here]

Flow canvas

Figure 7.2: Flow in Flow Designer

[Paste screenshot here]

Approved request

Figure 7.3: Request item with automatic approval

[Paste screenshot here]

Catalog task

Figure 7.4: Catalog task assigned to IT Procurement

[Paste screenshot here]

Notification / execution

Figure 7.5: Email notification and flow execution status

8. ADVANTAGES & DISADVANTAGES

Advantages

- Faster ordering with automatic approval of standard models.
- Low-code, visual flow that is easy to read and modify.
- Reduces manual effort and data re-entry for procurement staff.
- Consistent policy enforcement and full tracking of every order.
- Better experience through status emails.

Disadvantages

- Depends on the ServiceNow platform and its licences.
- Standard model list must be maintained manually.
- Does not yet connect to vendors or inventory for automatic purchase.
- Complex exceptions still need manual handling.

9. CONCLUSION

The project aimed to streamline IT procurement by automating standard laptop orders. A catalog item and a Flow Designer flow were created to validate requests, approve standard models automatically, route exceptions to managers, create fulfilment tasks, notify requesters and close requests. All planned test cases behaved as expected, so the objectives were achieved. The team learned how to design catalog items, build and debug flows, and apply low-code automation to a real business process.

10. FUTURE SCOPE

- Integrate with vendor or procurement systems to place orders automatically.
- Check inventory before ordering and offer available stock first.
- Extend the flow to monitors, phones and other standard hardware.
- Add a dashboard for order volume, delivery time and cost.
- Use a lookup table or property to manage the standard model list.
- Add mobile access and approval from email or chat.

11. APPENDIX

Source Code (if any)

No custom code was written; the solution is configured using low-code tools. Key configuration:

Component	Configuration
Catalog item	Standard Laptop Request (variables: laptop model, requested for, justification)
Flow name	Standard Laptop Order Automation
Trigger	Service Catalog - runs when the catalog item is submitted
Approval	Auto-approve if model is standard; otherwise manager approval
Fulfilment	Catalog task assigned to IT Procurement group
Notifications	Email to requester on approval, rejection and completion

Dataset Link

No external dataset was used. A sample list of standard laptop models and test requests were created manually in the ServiceNow instance.

GitHub & Project Demo Link

GitHub repository:
Project demo link:

Steps to reproduce: (1) Create the assignment group IT Procurement. (2) Create the catalog item and its variables. (3) Open Flow Designer and create a new flow with the Service Catalog trigger. (4) Add the actions and logic from the flow outline in Section 4.2. (5) Activate the flow and publish the catalog item. (6) Submit test requests as employee and manager users and review the Flow Execution view.